QP CODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION

OCTOBER 2025

Third Semester

DISCIPLINE SPECIFIC CORE COURSE B (DSC B)– MG3DSCGEO202

MINERAL SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer in 60 words

Answer any **7** questions

Each question carries **2** marks

1. Define polymorphism. [A][2]
2. Name any two non-silicate minerals. [A][4]
3. Define Solid solutions in minerals. [A][2]
4. List some important parts of petrological microscope. [A][3]
5. List the Scope of Optical Mineralogy. [A][3]
6. Define critical angle. [A][3]
7. List the Physical properties of Garnet Group of Minerals. [An][5]
8. Name any two Native Elements. [An][6]

[2x7 = 14]

Part B

Short notes in 250 words

Answer any **3** questions

Each question carries **8** marks

9. Classify Minerals based on anionic groups. [A][4]

10. Explain mode of occurrence and uses of the group oxides. [An][6]

11. Describe physical properties of carbonates, sulfates and halides. [An][6]

12. Illustrate and explain nicol prism. [A][3]

13. List the symmetry elements and forms present in the normal class of Tetragonal. [U][1]

Part C

Essay in 400 words

Answer any **1** questions

Each question carries **12** marks

14. Explain Optical properties under plane polarised and crossed polarised light. [A][3]

15.Explain Structure, chemistry and optical and physical properties, modes of occurrence of Phyllosilicates. [An][5]

[12x1 = 12]